Phylogenetic Super Tree Construction

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Phylogenetic Super Tree Construction

To estimate effects of nonnative species occurrences on phylogenetic distinctiveness of stream fishes, we constructed an informal, time calibrated, phylogenetic super tree containing all 815 species in the FishScales stream fish community dataset. Informal phylogenetic super trees are constructed of multiple smaller, often incomplete phylogenetic trees, that were potentially constructed using different methods (Bansal et al., 2010). Informal super trees represent untested phylogenetic hypotheses, and thus caution should be used when making evolutionary inferences with super trees (Castiglione et al., 2022). As we are interested only in phylogenetic diversity and not the testing of evolutionary hypotheses, super trees are adequate for capturing general phylogenetic relationships among species and addressing the goals of this study.

To construct the super tree, we used the methods of Castiglione et al. (2022) with the *tree.merger* function from the 'RRphylo' R package (Raia et al., 2018). In this manner, one tree acts as the backbone tree in which nodes and tips are added using the other trees as a reference for node position and age. We used the Fish Tree of Life (Chang et al., 2019; Rabosky et al., 2018), an extensive phylogeny of the class Actinopterygii, as the backbone tree. To ensure compatibility between the super tree and our community dataset, we harmonized the tree's tip labels to match those in dataset and coarsened species to subspecies. Upon harmonization, the tree contained most of the species in the full community data (~89%). Despite this large overlap with our data, there were entire groups (e.g., lampreys) and many species in the community dataset not present in the tree and as a result 89 species were added to the super tree using phylogenetic data from TimeTree 5 (Kumar et al., 2022) and other smaller taxon-specific trees.

TimeTree 5 is a public database which contains a time calibrated phylogenetic tree assembled from thousands of published studies.

The backbone tree only contained species from Actinopterygii, and as such did not include any lampreys (class Petromyzontida). However, the community dataset contained 12 species of lamprey, all of which were included in TimeTree 5 (Table 1). The lamprey reference tree was added as an outgroup to the backbone tree using the node age supplied by the reference tree (563.4 mya). The remainder of the missing species were part of Actinopterygii and could be added as ingroup clades to the backbone tree. When possible, we used time-calibrated phylogenetic trees as reference trees to add missing species to the backbone tree. Time calibrated trees allowed us to place tips using node ages from the reference tree to utilize as much phylogenetic data from the reference tree without altering the data in the backbone tree. We were able to add 26 species in this manner (Table 2). However, there were cases in which species could not be added into the backbone tree using the refence tree's node ages. These cases were when (1) the node ages in the reference tree altered the existing node ages in the backbone tree, (2) species had phylogenetic trees that were not time calibrated (e.g., phylograms and cladograms), and (3) species did not have any phylogenetic trees available.

One situation in which we could not add tips to the backbone tree with node ages from the reference tree occurred when the reference node ages did not fit into the existing backbone tree structure. This would occur when the node age for an added tip predated the node it was being added under, which would not alter the tree height, but result in many of the nodes above this point to be recalibrated. A similar situation occurred when the node age for an added tip was younger than the node it was being added above; this would cause all nodes below this point to be recalibrated. To maintain the structure and node ages of the backbone tree, we first attempted to replace the entire group of tips and nodes in disagreement with the reference tree. We were able to replace the entire *Campostoma* and *Micropterus* genera in the backbone tree using the references trees to add missing species with reference node ages (Table 3). When swapping out larger portion of the trees did not fix the recalibration issue, we add the missing species' tips without suppling node ages. Here, *tree.merger* places the node as such to not alter existing node ages by placing the new node in between the node above and the node/tip below the addition. We added 8 species in this manner (Table 4).

Another situation in which we were not able to assign the reference node ages to new tips, was when we were only able to find phylogenetic trees that were not time calibrated for those missing species. This was the case for 19 species (Table 5). As we could only infer relationship from these species, the tips for these species were added using node ages assigned by *tree.merger* as described in the above section.

Finally, we were not able to assign reference node ages to some tips, as we could not find phylogenetic trees for those species. There were 16 species in the community dataset that we could not find trees for (Table 6). These species were placed in the backbone tree as sister species to closely related species of the same genus based on knowledge from existing literature (Beaulieu et al., 2012). Species without phylogenetic trees were mainly provisional species, rare species, or species resulting from recent taxonomic splits. For provisional species, tips were added as a sister to the species that the provisional species is described as being related (e.g., *Notropis sp. cf. chlorocephalus* was added as a sister to *Notropis chlorocephalus*). Rare species were added sister to species that are described as being closely related to that species. Finally, tips for species resulting from a taxonomic split were added as a sister to the species they were originally described as. All of the above cases were added to the backbone tree using node ages assigned by *tree.merger*.

Once all missing species were added to the backbone tree, we visually inspected the tree structure and node ages to ensure all tips and nodes were added to the tree correctly.

Table 1 Table showing the name of lamprey species added as an outgroup to the backbone tree using node ages from reference tree. The reference tree citation is listed for the reference trees used for tip placement and node ages for each species (all were TimeTree 5).

Tip label	Reference tree citation
Entosphenus tridentatus	TimeTree 5
Ichthyomyzon bdellium	TimeTree 5
Ichthyomyzon castaneus	TimeTree 5
Ichthyomyzon fossor	TimeTree 5
Ichthyomyzon gagei	TimeTree 5
Ichthyomyzon greeleyi	TimeTree 5
Ichthyomyzon unicuspis	TimeTree 5
Lampetra aepyptera	TimeTree 5
Lampetra ayresii	TimeTree 5
Lampetra hubbsi	TimeTree 5
Lampetra richardsoni	TimeTree 5
Lethenteron appendix	TimeTree 5
Petromyzon marinus	TimeTree 5

Table 2 Table showing the name of species added to backbone tree using node ages from reference tree. The reference tree citation is listed for the reference trees used for tip placement and node ages for each species.

Tip label	Reference tree citation
Catostomus latipinnis	Bagley et al. (2018)
Labidesthes vanhyningi	Bloom et al. (2009)
Macrhybopsis boschungi	Hoagstrom & Echelle (2022)
Macrhybopsis etnieri	Hoagstrom & Echelle (2022)
Macrhybopsis pallida	Hoagstrom & Echelle (2022)
Macrhybopsis tetranema	Hoagstrom & Echelle (2022)
Notropis amplamala	Hollingsworth et al. (2013)
Erimyzon claviformis	Hunt et al. (2021)
Lepomis peltastes	Kim, Bauer, et al. (2022)
Chrosomus neogaeus	TimeTree 5
Coregonus artedi	TimeTree 5
Ctenogobius shufeldti	TimeTree 5
Erimonax monachus	TimeTree 5
Gobioides broussonetii	TimeTree 5
Hypostomus plecostomus	TimeTree 5
Lepidomeda copei	TimeTree 5
Microgobius gulosus	TimeTree 5
Neogobius melanostomus	TimeTree 5
Notropis alborus	TimeTree 5
Notropis bairdi	TimeTree 5
Notropis braytoni	TimeTree 5
Notropis cummingsae	TimeTree 5
Notropis scabriceps	TimeTree 5
Poecilia formosa	TimeTree 5
Pteronotropis merlini	TimeTree 5

Table 3 Table of species added to backbone tree by replacing an entire genus. Species with a value of TRUE for *Present in backbone* column were species that were in the backbone tree but needed to be replaced with data from reference tree as to not alter the existing branch lengths in backbone tree. The reference tree citation is listed for the reference trees used for tip placement and node ages for each species. Provisional species are noted with the format: *Genus sp. common name*.

Tip Label	Present in backbone	Reference tree citation
Campostoma anomalum	TRUE	Hollingsworth et al. (2013)
Campostoma oligolepis	TRUE	Hollingsworth et al. (2013)
Campostoma ornatum	TRUE	Hollingsworth et al. (2013)
Campostoma pauciradii	TRUE	Hollingsworth et al. (2013)
Campostoma spadiceum	FALSE	Hollingsworth et al. (2013)
Micropterus cataractae	TRUE	Kim, Taylor et al. (2022)
Micropterus chattahoochae	FALSE	Kim, Taylor et al. (2022)
Micropterus coosae	TRUE	Kim, Taylor et al. (2022)
Micropterus dolomieu	TRUE	Kim, Taylor et al. (2022)
Micropterus henshalli	TRUE	Kim, Taylor et al. (2022)
Micropterus punctulatus	TRUE	Kim, Taylor et al. (2022)
Micropterus salmoides	TRUE	Kim, Taylor et al. (2022)
Micropterus treculii	TRUE	Kim, Taylor et al. (2022)
Micropterus cahabae	FALSE	Kim, Taylor et al. (2022)
Micropterus sp. Altamaha Bass	FALSE	Kim, Taylor et al. (2022)
Micropterus sp. Bartram's Bass	FALSE	Kim, Taylor et al. (2022)
Micropterus sp. Choctaw Bass	FALSE	Kim, Taylor et al. (2022)
Micropterus tallapoosae	FALSE	Kim, Taylor et al. (2022)
Micropterus warriorensis	FALSE	Kim, Taylor et al. (2022)

Table 4 Table showing the name of species added to backbone tree without using node ages from reference tree. These tips/nodes were added using node ages assigned by the *tree.merger* function. The reference tree citation is listed for the reference trees used for tip placement for each species. Provisional species are noted with the format: *Genus sp. common name*.

Tip label	Reference tree citation
Catostomus fumeiventris	Bagley et al. (2018)
Moxostoma sp. Apalachicola Redhorse	Bagley et al. (2018)
Moxostoma sp. Brassy Jumprock	Bagley et al. (2018)
Moxostoma sp. Sicklefin Redhorse	Bagley et al. (2018)
Leptolucania ommata	TimeTree 5
Notropis atrocaudalis	TimeTree 5
Notropis greenei	TimeTree 5
Notropis melanostomus	TimeTree 5

Table 5 Table showing the name of species added to backbone tree from non-time calibrated trees (e.g., phylograms and cladograms). These tips/nodes were added using node ages assigned by the *tree.merger* function. The reference tree citation is listed for the reference trees used for tip placement for each species. Provisional species are noted with the format: *Genus sp. common name*.

Tip label	Reference tree citation
Notropis sp. Sawfin Shiner	Hollingsworth et al. (2013)
Cottus kanawhae	Kinziger et al. (2005)
Cottus sp. Bluestone Sculpin	Kinziger et al. (2005)
Cottus sp. Checkered Sculpin	Kinziger et al. (2005)
Cottus sp. Clinch Sculpin	Kinziger et al. (2005)
Cottus sp. Holston Sculpin	Kinziger et al. (2005)
Etheostoma gore	Layman & Mayden (2012)
Etheostoma teddyroosevelt	Layman & Mayden (2012)
Pteronotropis stonei	Mayden & Allen (2015)
Etheostoma sp. Blueface Darter	Near et al. (2011)
Etheostoma atripinne	Near et al. (2011)
Etheostoma gutselli	Near et al. (2011)
Etheostoma jimmycarter	Near et al. (2011)
Etheostoma sp. Clarks Darter	Near et al. (2011)
Etheostoma sp. Mamequit Darter	Near et al. (2011)
Etheostoma sp. Sipsey Darter	Near et al. (2011)
Etheostoma trisella	Near et al. (2011)
Elassoma okatie	Sandel et al. (2014)
Cyprinella sp. Thinlip Chub	Schönhuth & Mayden (2010)

Table 6 Table showing the name of species added to backbone tree that did not have a phylogenetic tree available. These species were either provisional species, rare species, or species resulting from taxonomic splits. Provisional species are noted with the format: *Genus sp. common name*. These tips/nodes were added as sister species to existing taxa (tip location) based on existing literature (justification). Node ages assigned by the *tree.merger* function.

Tip label	Type	Tip location	Justification
Clinostomus sp. Smoky Dace	Provisional species	*Clinostomus funduloides*	Described as *Clinostomus sp. cf. funduloides*
Cottus sp. Colorado River Sculpin	Provisional species	*Cottus annae*	Described as being related to *Cottus annae* (Young et al., 2022)
Cottus sp. Columbia Slimy Sculpin	Provisional species	*Cottus cognata*	Described as *Cottus sp. cf. cognata*
Cottus sp. Rocky Mountain Sculpin	Provisional species	*Cottus bairdii*	Described as *Cottus sp. Cf bairdii*
Moxostoma sp. Carolina Redhorse	Provisional species	*Moxostoma erythrurum*	Described as *Moxostoma sp. cf. erythrurum*
Notropis sp. Kanawha Rosyface Shiner	Provisional species	*Notropis rubellus*	Described as *Notropis sp. Cf. rubellus*
Notropis sp. Piedmont Shiner	Provisional species	*Notropis chlorocephalus*	Described as *Notropis sp. cf. chlorocephalus*
Noturus sp. Highlands Stonecat	Provisional species	*Noturus flavus*	Described as *Noturus sp. cf. flavus*
Macrhybopsis australis	Rare species	*Macrhybopsis tetranema*	Described as sister species to *Macrhybopsis tetranema* (Underwood et al., 2003)
Margariscus nachtriebi	Rare species	*Margariscus margarita*	*Margariscus margarita* is the only other species in this genus
Notropis albizonatus	Rare species	*Notropis procne*	Described as a member of the *Notropis procne* species group (Warren et al., 1994)
Catostomus utawana	Taxonomic split	*Catostomus commersonii*	Formerly described as *Catostomus commersonii.* (Morse & Daniels, 2009)
Cottus immaculatus	Taxonomic split	*Cottus hypselurus*	Formerly described as *Cottus hypselurus* (Kinziger & Wood, 2010)
Menidia audens	Taxonomic split	*Menidia beryllina*	Formerly described as *Menidia beryllina* (Fluker et al., 2011)
Notropis buccula	Taxonomic split	*Notropis bairdi*	Formerly described *as Notropis bairdi* (Page & Burr, 1991)

Oncorhynchus aguabonita	Taxonomic split	*Oncorhynchus mykiss*	Formerly described as *Oncorhynchus mykiss* (NatureServe, 2010)

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